



This sample of neon is stored in a glass sphere. Normally a colorless gas, it gives off a red-orange glow when electrified.

Neon

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Perhaps the most well known of all noble gases for its role in neon signs, neon is an unreactive element. Like helium, this transparent gas is commonly found throughout the universe but is rare on Earth. Unlike helium, which is found mixed in natural gas, neon is available only in air. It is extracted by cooling air until it becomes a liquid. In industry, neon is used as a coolant as well as in making some types of fluorescent lamps and in diving equipment.



Neon lights

Gas tubes filled with neon are commonly used in bright red signs and lights. Although they are all called "neon lights," the other colors are not actually neon, but other elements—for example, helium (yellow), argon (purple), and mercury (blue).



A helium-neon laser beam appears red in color.

Scanning codes

Bar codes store data about a product that can only be read by machines. Some bar code scanners use lasers made of a combination of helium and neon. These efficient lasers use very little energy and are used in supermarkets and retail shops throughout the world.

Weather balloons collect wind data, including temperature and speed, as well as atmospheric pressure readings at high altitudes.

Up, up, and away

Helium has been used in all types of balloons, including weather balloons and party balloons. Since helium gas is lighter than air, balloons filled with it float upward. Unlike hydrogen, helium is not flammable, making it a much safer choice.

